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Madan Baduwal

Spectrum Sharing, Wireless Communications, Machine Learning, Robotics

Education

Exp. Dec 2028 **PhD in Computer Science & Engineering**, *Mississippi State University*, Starkville, MS.
GPA: 4.0/4.0
Advisor: Dr. Vini Chaudhary
Research: Machine Learning, Wireless Communications, Spectrum Sharing

Dec 2024 **MS in Computer Science**, *University of Texas Permian Basin*, Odessa, TX.
GPA: 4.0/4.0
Advisor: Dr. Quan Yuan
Research: Machine Learning, Computer Networks, Wireless Networks

Dec 2020 **B.E. in Computer Engineering**, *Tribhuvan University*, Kathmandu, Nepal.
GPA: 72.38/100
Relevant Coursework: Electromagnetics, Digital Signal Analysis and Processing, Mathematics (Set Theory, Linear Algebra, Calculus, probability & Statistics), Data Structures & Algorithms, Machine Learning, Computer Networks, Data Communication

Research Experience

Jan 2025 – Present **Graduate Research Assistant**, *Mississippi State University*, Starkville, MS, USA.
- Developed multimodal learning methods for radar interference detection in CBRS under challenging low-SINR conditions.
- Designed federated and personalized learning approaches for distributed radar detection across heterogeneous wireless environments.
- Developed OOD detection and continual learning strategies for identifying unseen spectrum conditions and adapting models over time.
- Designed lightweight and communication-efficient learning methods for wireless and edge intelligence.
- Developing quantization-aware multimodal federated learning methods for communication-efficient distributed spectrum sensing.

Jan 2023 – Dec 2024 **Graduate Research Assistant**, *University of Texas Permian Basin*, Odessa, TX, USA.
- Conducted machine learning research in medical image segmentation and multimodal learning, including optimizing and fine-tuning models such as YOLOv8, U-Net, Detectron, SAM, Mask R-CNN, OneFormer, CLIP, BLIP, LLaVA, ViLBERT, and LXMERT.
- Conducted research on the applications of machine learning in computer networking, examining current challenges, applications, and future research directions.
- Conducted research on machine learning approaches for phishing website detection, analyzing existing learning paradigms and techniques for identifying malicious websites.

Mar 2023 – Dec 2024 **Student Assistant Editor**, *University of Texas Permian Basin*, Odessa, TX, USA.
- Refined the editing and proofreading of the UTPB Journal of Undergraduate Research, Volume 5, 2023, and the UG Research Book 2024 for publication under the direction of Prof. Rebecca Babcock and Prof. Mohamed K Zobaa.

Published/Accepted Papers

M. Baduwal and V. Chaudhary, "Lightweight Multimodal Radar Interference Detection at Low SINRs in CBRS," in *IEEE International Conference on Communications (ICC)*, 2026, pp. 1–6.

S. U. Khan, **M. Baduwal**, V. Chaudhary, and D. Roy, "PERFECT: Personalized Federated Learning for CBRS Radar Detection," in *35th International Conference on Computer Communications and Networks (ICCCN)*, 2026, pp. 1–6.

W. Patterson, **M. Baduwal**, and V. Chaudhary, "Who Ran My Circuit: Calibration Data-Based Fingerprinting of Quantum Cloud Hardware," in *35th International Conference on Computer Communications and Networks (ICCCN)*, 2026, pp. 1–6.

M. Baduwal, P. Paudel, and V. Chaudhary, "Federated Learning: A Survey of Core Challenges, Current Methods, and Opportunities," *Computers*, vol. 15, no. 3, Art. no. 155, 2026. **[Featured as Cover Story Article]**

P. R. Prajapati, S. Poudel, **M. Baduwal**, S. Burlakoti, and S. P. Panday, "Signature Verification using Convolutional Neural Network and Autoencoder," *Journal of the Institute of Engineering*, vol. 16, no. 1, 2021.

Preprints

M. Baduwal and P. Paudel, "Radar Detection in the CBRS Band: Techniques, Challenges, and Future Directions," *arXiv preprint arXiv:2608.01786*, 2026.

M. Baduwal, "Hybrid (Transformer+CNN)-based Polyp Segmentation," *arXiv preprint arXiv:2508.09189*, 2025.

M. Baduwal and P. Paudel, "QFedPolyp: A Communication- and Inference-Efficient Federated Learning Framework for Polyp Segmentation," *arXiv preprint arXiv:2607.22743*, 2026.

P. Paudel and **M. Baduwal**, "Benchmarking Machine Learning Models for Multi-Omics-Based Breast Cancer Prediction," *arXiv preprint arXiv:2607.16250*, 2026.

M. Baduwal and P. Paudel, "Evaluating Agentic Artificial Intelligence: A Comprehensive Survey of Metrics, Benchmarks, and Methodologies," *TechRxiv preprint*, 2026.

Awards and Achievements

Apr 2026 **NSF Student Travel Grant** (IEEE ICC 2026). Awarded $1,538 NSF-funded travel grant (13 selected from 42 applicants) to present research in Glasgow, UK.

Feb 2026 **Travel Assistance Grant for Graduate Students (TAGGS)** (Mississippi State University). Awarded $800 competitive travel grant for conference presentation.

Feb 2026 **Dean's Travel Award** (Bagley College of Engineering, Mississippi State University). Received $500 funding for IEEE ICC 2026 presentation.

2026 **Departmental Travel Support** (Computer Science & Engineering, Mississippi State University). Awarded $200 funding for international conference travel.

Aug 2015 – Dec 2019 **Merit-Based Scholarship** (Tribhuvan University). Awarded $3,000 for highest GPA performance (3 of 7 semesters).

Jul 2019 **Best Logic Code Award** (Sagarmatha Engineering College). Secured 1st place and received $600 prize.

Mar 2018 **Hackathon Runner-Up** (Kathmandu University). Awarded 1st runner-up position with $1,000 prize.

Jan 2017 **Best Idea Award** (Kantipur Engineering College). Won best idea award with professional training prize (approximately $200 value).

Academic Service

Peer Reviewer Reviewed submissions for major IEEE conferences including: **COMSNETS**, **IEEE CCNC**, **IEEE ICNC**, and **IEEE ICCCN** (Total: 17+ papers).

TPC Member **IEEE CCNC 2027** - Tracks: Edge/Cloud Computing and Networking; Security, Privacy, and Content Protection.

Research Evaluator Undergraduate Research Symposium, Mississippi State University, 2026

Industry Experience

Jan 2022 – Dec 2022 **Senior Machine Learning Engineer**, *Fusemachines*, Kathmandu, Nepal.

- Led, designed, and developed a no-code data-centric AI platform (Matrice.ai) for building and deploying ML apps, reducing user deployment time by 40% and development costs by 80% using PyTorch, React, Django, and AWS.
- Developed an ML app as a SaaS product for 5+ hospitals, using PyTorch, Django, CI/CD, and Dell EMC servers.
- Optimized and automated instance segmentation (accuracy 85%) for otitis media Otoscopy images of the tympanic membrane using CNN, driving $100k of revenue annually by increasing the doctor-patient ratio by 1.5x.

Feb 2020 – Dec 2021 **Machine Learning Engineer**, *Fusemachines*, Kathmandu, Nepal.

- Led a 10-person team to develop an AI-enabled student progress tracking platform (fuseclassroom.com), increasing the teacher-student ratio by 2x. Currently operational in ∼60 Nepalese colleges with 20k active students.
- Orchestrated a phishing detection engine for over 10 clients under WMC Global, improving accuracy from 85% to 90% by selecting the optimal number of clusters using the silhouette score, detecting millions of URLs per day.
- Delivered over 20+ trainings, workshops, knowledge-sharing, and paper reading sessions on mathematics and machine learning. Empowered the ML skills of ∼5k students by collaborating with schools, colleges, and universities.
- Enriched affordable AI courses by researching, designing, reviewing, and refining content, including reading material, quizzes, assignments, and projects for Fusemachines AI Education Programs, focusing on AI, ML, CV, and NLP.

Jan 2021 – Dec 2021 **Computer Vision Engineer (R&D), Part-Time**, *National Innovation Center*, Kathmandu, Nepal.

- Led and collaborated with mechanical, electrical, and electronic hardware teams to deploy computer vision tasks in robots, resulting in the creation of initial prototypes for waiter and service robots within 9 months.
- Achieved 60% inference acceleration on computer vision tasks integrating into robots using edge-computing platforms and hardware accelerators, including Google Coral TPU, NVIDIA Jetson Nano, and Raspberry Pi, with cloud integration through Microsoft Azure.
- Implemented and fine-tuned computer vision tasks, including object detection, segmentation, tracking, face recognition, and depth estimation, using PyTorch and the Intel RealSense D435i depth camera, covering the entire vision pipeline.
- Improved pathfinding algorithms (A* and Dijkstra), leading to a 20% reduction in execution time and faster robotic navigation.

Aug 2019 – Feb 2020 **Software Engineer Intern**, *Omnibluetech*, Kathmandu, Nepal.

- Devised and architected Django REST APIs and AWS-based background workers capable of managing millions of daily requests, facilitating third-party entities in extracting data from unstructured documents in a scalable fashion.
- Engineered web applications using front-end and back-end technologies: HTML5, CSS, JavaScript, and Django for retail stores and consultancies, resulting in a 15% increase in revenue for the company in the first quarter.

Projects

Jun 2021 – Feb 2022 **Phishing Detection** | Python 3, Scikit-learn, Git, GitHub, AWS, Docker, Kubernetes, CI/CD.

- Classified websites into phishing and non-phishing categories using deep learning algorithms for GoDaddy, Bitly, InfoBip, and ICANN under WMC Global. Transformed unstructured data into structured data format using RegEx.

Jan 2021 – Jun 2021 **Student Status Engine** | Python 3, Snowflake, Scikit-learn, Git, AWS.

- Crafted a feature extractor pipeline that automatically extracts features from the data warehouse (e.g., Snowflake). Leveraged machine learning algorithms on extracted features to classify student status into different classes.
- Collaborated with back-end, front-end, and DevOps engineers to test and deploy a model into large-scale production.

Jan 2021 – Dec 2021 **Mina** (AI-Robot) | Python 3, PyTorch, ROS, Gazebo, Coral TPU, Jetson Nano, Raspberry Pi.
- Retrained a classification model for Edge TPU using post-training quantization (23 FPS and 85% mAP score with pre-training), face recognition using the Python 3 face recognition library, depth calculation using a RealSense depth camera, Centroid-based object tracking, and wrote a rule-based algorithm from scratch.
- Incorporated and validated computer vision tasks into the ROS and Gazebo simulation environments and implemented them into waiter and service robots. Visualized robot sensor data using RViz.

Jan 2020 – Oct 2020 **Text Extractor** | Jupyter Notebook, Python 3, OpenCV.
- Conducted research and experiments on building image preprocessing techniques like erosion and dilation. Built a framework that uses Google Tesseract and RegEx to extract information from the form (e.g., buyer name, seller name, etc.).

Jan 2017 – Dec 2019 **Hastakshar** | Python 3, OpenCV, Keras, Django.
- Performed research and experimentation on image localization to boost classifier model accuracy to 83% with NumPy and OpenCV.
- Built a signature verification CNN classifier system using TensorFlow and Django web interfaces on local machines.

2015 – 2019 **Android apps and websites** | C#, Unity, Python 3, HTML, CSS.
- Android apps: Asteroid Smash, Antigravity Ball, Saveme, Beat Creator, and 1K downloads
- Web apps: horizonglobal.edu.np, youthcareer.edu.np: between 10,000 and 15,000 visitors per month

Skills

Languages	Python, R, Julia, C++, C, Java, JavaScript, C#, Go, Kotlin, Rust, Bash, MATLAB, SQL, LaTeX
Machine Learning	PyTorch, TensorFlow, JAX, Scikit-learn, NumPy, Pandas, OpenCV, NLTK, Hugging Face, Transformers, CNNs, GNNs, Diffusion Models
Data Engineering	PostgreSQL, MySQL, MongoDB, Snowflake, Apache Spark, Hadoop, Kafka, Databricks, ETL Pipelines, Tableau, Power BI
Frameworks	Django, Flask, Node.js, React, Angular, React Native, Streamlit, REST APIs, ROS, Unity
DevOps	AWS, GCP, Azure, Docker, Kubernetes, Jenkins, Airflow, MLflow, DVC, SageMaker, Git, GitHub
Other Tools	Redis, RabbitMQ, Elasticsearch, Ansible, Datadog, Agile, Jira, Confluence